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APUL

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Exercise - I Basic Mathematics Used in Physics and Vectors

1. The moon's distance from the earth is 360000 km and its diameter subtends an angle of 42' at the eye of the observer. The diameter of the moon is

(1) 4400 km (2) 1000 km
(3) 3600 km (4) 8800 km

PBM236

2. A unit radial vector $\hat{r}$ makes angles of $\alpha = 30°$ relative to the x-axis, $\beta = 60°$ relative to the y-axis, and $\gamma = 90°$ relative to the z-axis. The vector $\hat{r}$ can be written as :

(1) $\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}\hat{j}$ (2) $\frac{\sqrt{3}}{2}\hat{i}+\frac{1}{2}\hat{j}$
(3) $\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}$ (4) None of these

PBM237

3. If velocity of a particle is given by v = (2t + 3) m/s, then average velocity in interval $0 \le t \le 1s$ is :

(1) $\frac{7}{2}$ m/s (2) $\frac{9}{2}$ m/s
(3) 4 m/s (4) 5 m/s

PBM238

4. Square of the resultant of two forces of equal magnitude is equal to three times the product of their magnitude. The angle between them is :

(1) 0° (2) 45° (3) 60° (4) 90°

PBM239

5. Given that $\vec{A}+\vec{B}+\vec{C}=\vec{0}$. Out of these three vectors two are equal in magnitude and the magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. Then the angles between vectors are :

(1) 30°, 60°, 90° (2) 45°, 45°, 90°
(3) 45°, 60°, 90° (4) 90°, 135°, 135°

PBM240

6. A vector of length ℓ is turned through the angle θ about its tail. What is the change in the position vector of its head?

(1) $\ell \cos(\theta/2)$ (2) $2\ell \sin(\theta/2)$
(3) $2\ell \cos(\theta/2)$ (4) $\ell \sin(\theta/2)$

PBM241

7. Vector $\vec{a}$ is turned without a change in its length through a small angle 'dθ'. The value of $|\Delta\vec{a}|$ and Δa are respectively.

(1) 0, adθ (2) adθ, 0
(3) 0, 0 (4) None of these

PBM242

8. The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If $\vec{Q}$ is doubled then the new resultant vector is perpendicular to '$\vec{P}$'. Then R is equal to :

(1) $\left(\frac{P^2-Q^2}{2PQ}\right)$ (2) Q
(3) $\frac{P}{Q}$ (4) $\frac{P+Q}{P-Q}$

PBM243

9. $\vec{A}$ and $\vec{B}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{A}+\vec{B}|+|\vec{A}-\vec{B}|$ is :-

(1) 2 (2) 0
(3) 4 (4) $\sqrt{3}+1$

PBM244

10. If $\vec{A}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{B}=-\hat{i}+\hat{j}+4\hat{k}$ and $\vec{C}=3\hat{i}-3\hat{j}-12\hat{k}$ then find the angle between the vector $(\vec{A}+\vec{B}+\vec{C})$ and $(\vec{A}\times\vec{B})$ in degrees.

(1) 90° (2) 45°
(3) 0° (4) 180°

PBM245

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	2	3	3	4	2	2	2	3	1

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Exercise - II Basic Mathematics Used in Physics and Vectors

1. At what angle must the two forces (x + y) and (x – y) act so that the resultant may be $\sqrt{(x^2+y^2)}$?

(1) $\cos^{-1}\left[\frac{-(x^2+y^2)}{2(x^2-y^2)}\right]$

(2) $\cos^{-1}\left[\frac{-2(x^2-y^2)}{x^2+y^2}\right]$

(3) $\cos^{-1}\left[\frac{-(x^2+y^2)}{x^2-y^2}\right]$

(4) $\cos^{-1}\left[\frac{(x^2-y^2)}{x^2+y^2}\right]$

PBM246

2. Given that P = Q = R. If $\vec{P}+\vec{Q}=\vec{R}$ then the angle between $\vec{P}$ & $\vec{R}$ is θ_1. If $\vec{P}+\vec{Q}+\vec{R}=\vec{0}$ then the angle between $\vec{P}$ & $\vec{R}$ is θ_2. What is the relation between θ_1 and θ_2?

(1) $\theta_1 = \theta_2$

(2) $\theta_1 = \frac{\theta_2}{2}$

(3) $\theta_1 = 2\theta_2$

(4) None of the above

PBM247

3. The sum of magnitudes of two forces acting at a point is 16N. If the resultant force is 8N and its direction is perpendicular to smaller force, then the forces are :

(1) 6N & 10N

(2) 8N & 8N

(3) 4N & 12N

(4) 2N & 14N

PBM248

4. Find the parallel and perpendicular components of vector $(2\hat{i}+3\hat{j})$ to $(3\hat{i}+4\hat{j})$

(1) $\frac{1}{2}(\hat{j}-\hat{i}),\frac{7}{2}(\hat{j}+\hat{i})$

(2) $\frac{3}{2}(\hat{j}+\hat{i}),\frac{5}{2}(\hat{j}+\hat{i})$

(3) $\frac{54\hat{i}+72\hat{j}}{25},\frac{-4\hat{i}+3\hat{j}}{25}$

(4) $\frac{\hat{i}-2i}{\sqrt{2}},\frac{\hat{i}-\hat{j}}{2}$

PBM249

Exercise-II ANSWER KEY

Question	1	2	3	4
Answer	1	2	1	3

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PBM249

Exercise-II ANSWER KEY

Question	1	2	3	4
Answer	1	2	1	3

Exercise - I Kinematics

1. In a straight line motion the distance travelled is proportional to the square root of the time taken. The acceleration of the particle is proportional to : -
(1) v (2) v^2 (3) v^3 (4) $\sqrt{v}$
PMP282

2. A man throws balls with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given g = 9.8 m/s²)
(1) More than 19.6 m/s.
(2) At least 9.8 m/s.
(3) Any speed less than 19.6 m/s.
(4) Only with speed 19.6 m/s.
PMP283

3. A car is moving with speed u. Driver of the car sees red traffic light. His reaction time is t, then find out the distance travelled by the car after the instant when the driver decided to apply brakes. Assume uniform retardation 'a' after applying brakes.
(1) $ut+\frac{u^2}{a}$ (2) $ut+\frac{u^2}{2a}$
(3) $2ut+\frac{u^2}{a}$ (4) $2ut+\frac{u^2}{2a}$
PMP284

4. Three different objects of masses m_1, m_2 and m_3 are allowed to fall from rest and from the same point 'O' along three different frictionless paths. The speeds of the three objects on reaching the ground, will be in the ratio of :–
(1) $m_1 : m_2 : m_3$ (2) $m_1 : 2m_2 : 3m_3$
(3) 1 : 1 : 1 (4) $\frac{1}{m_1} : \frac{1}{m_2} : \frac{1}{m_3}$
PMP285

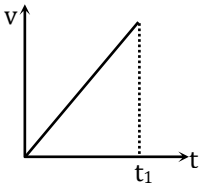
5. A body is thrown vertically upwards. If air resistance is to be taken into account, then the time during which the body rises is
(1) Equal to the time of fall
(2) Less than the time of fall
(3) Greater than the time of fall
(4) Twice the time of fall
PMP286

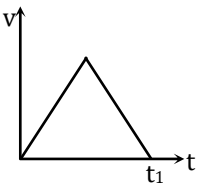
6. A particle is projected upwards. The times corresponding to height h while ascending and while descending are t_1 and t_2 respectively. The velocity of projection will be :
(1) gt_1 (2) gt_2
(3) $g(t_1 + t_2)$ (4) $\frac{g(t_1+t_2)}{2}$
PMP287

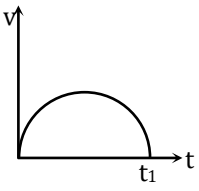
7. A particle is dropped from the top of a tower. The distance covered by it in the last one second is equal to that covered by it in the first three seconds. Find the height of the tower.
(1) 115 m (2) 125 m
(3) 135 m (4) 145 m
PMP288

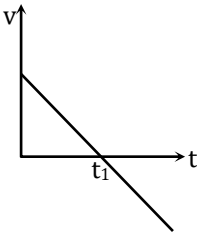
8. An elevator is accelerating upward at a rate of 6 ft/sec² when a bolt from its ceiling falls to the floor of the lift (Distance = 19 feet). The time (in seconds) taken by the falling bolt to hit the floor is (take g = 32 ft/ sec²)
(1) $\sqrt{2}$ (2) 1 (3) $2\sqrt{2}$ (4) $\frac{1}{2\sqrt{2}}$
PMP289

9. A ball is thrown straight up with velocity at t = 0 and returns to earth at t = t_1. Which graph shows the correct motion?

(1) 

(2) 

(3) 

(4) 

PMP290

10. A bomb is fired from a cannon with a velocity of 1000 m/s making an angle of 30° with the horizontal. What is the time taken by the bomb to reach at the highest point-
(1) 11 sec (2) 23 sec
(3) 38 sec (4) 50 sec
PMP291

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11. A projectile is projected with initial velocity $(5\hat{i}+12\hat{j})$ m/s. If g = 10 ms^{-2}, then horizontal range is :
(1) 4.8 metre (2) 9.6 metre
(3) 19.2 metre (4) 12 metre
PMP292

12. Two bodies of masses 100 kg and 50 kg are projected horizontally from same height with speeds 40 m/s and 20 m/s, simultaneously. The ratio of time taken by both the bodies to reach the ground is :-
(1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 1 : 4
PMP293

13. A ball is projected from the top of a tower at an angle 60° with the vertical. What happens to the vertical component of its velocity?
(1) Increases continuously
(2) Decreases continuously
(3) Remains unchanged
(4) First decreases and then increases
PMP294

14. A man walks on a straight road from his home to a market 2.5 km away with a speed of 5 km/h. Finding the market closed, he instantly turns and walks back home with a speed of 7.5 km/h. The average speed of the man over the interval of time 0 to 40 min is equal to
(1) 5 km/h (2) 25/4 km/h
(3) 30/4 km/h (4) 45/8 km/h
PMP295

15. A train is moving towards east and a car is along north, both with same speed. The observed direction of car to the passenger in the train is:-
(1) North–east direction
(2) North–west direction
(3) South-east direction
(4) None of these
PMP296

16. A stone is thrown upwards and it rises to a height of 100 m. The relative velocity of the stone with respect to earth will be maximum at height: -
(1) 100 m (2) 50 m
(3) Ground (4) 25 m
PMP297

17. If rain is falling at some angle from vertical and has horizontal velocity 2 m/s in east direction. With what velocity a man must move on the horizontal surface so that rain will appear vertical to him :-
(1) 4 m/s in east direction
(2) 2 m/s in east direction
(3) 2 m/s in west direction
(4) 2 m/s in a circular path
PMP298

18. A boy is running on a levelled road with velocity (u) with a long hollow tube in his hand. Water is falling vertically downwards with velocity (v). At what angle to the vertical, should he incline the tube so that the water drops enters without touching its side :
(1) $\tan^{-1}\left(\frac{v}{u}\right)$ (2) $\sin^{-1}\left(\frac{v}{u}\right)$
(3) $\tan^{-1}\left(\frac{u}{v}\right)$ (4) $\cos^{-1}\left(\frac{v}{u}\right)$
PMP299

19. A river is flowing at the rate of 8 km/h. A swimmer swims across the river with a velocity of 10 km/h w.r.t. water. The resultant velocity of the man will be in (km/h):-
(1) $\sqrt{117}$ (2) $\sqrt{340}$
(3) $\sqrt{164}$ (4) $3\sqrt{40}$
PMP300

20. A man wishes to swim across a river 0.5 km wide. If he can swim at the rate of $\sqrt{2}$ km/h in still water and the river flows at the rate of 1 km/h. The angle made by the direction (w.r.t. the flow of the river) along which he should swim so as to reach a point exactly opposite his starting point, should be :
(1) 60° (2) 120° (3) 135° (4) 90°
PMP301

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	2	3	2	4	2	2	4	4	4	1	4	4	2
Question	16	17	18	19	20										
Answer	3	2	3	3	3										

Exercise - II Kinematics

1. Position of a particle moving along x–axis is given by $x = 2 + 8t - 4t^2$. The distance travelled by the particle from t = 0 to t = 2 is :

(1) 0 (2) 8 (3) 12 (4) 16

PMP302

2. A stone is dropped from a height h. Another stone is thrown up simultaneously from the ground which reaches a height 4h. The two stones will cross each other after time:-

(1) $\sqrt{\frac{h}{8g}}$ (2) $\sqrt{8gh}$

(3) $\sqrt{2gh}$ (4) $\sqrt{\frac{h}{2g}}$

PMP303

3. A body is projected upwards with a velocity u. It passes through a certain point above the ground after t_1 second. The time after which the body passes through the same point during the return journey is:

(1) $\left(\frac{u}{g} - t_1^2\right)$ (2) $2\left(\frac{u}{g} - t_1\right)$

(3) $3\left(\frac{u^2}{g} - t_1\right)$ (4) $3\left(\frac{u^2}{g^2} - t_1\right)$

PMP304

4. A stone is dropped from the top of a tower of height h. After 1 second another stone is dropped from the balcony 20 m below the top of the tower. Both reach the bottom simultaneously. What is the value of h? take $g = 10\ m/s^2$:

(1) 3125 m (2) 312.5 m

(3) 31.25 m (4) 25.31 m

PMP305

5. A particle starts from rests and travels a distance S with uniform acceleration; then it travels a distance 2S with uniform speed; finally it travels a distance 3S with uniform retardation and comes to rest. If the complete motion of the particle is a straight line then the ratio of its average velocity to maximum velocity is :

(1) $\frac{6}{7}$ (2) $\frac{4}{5}$ (3) $\frac{3}{5}$ (4) $\frac{2}{5}$

PMP306

6. A body is thrown vertically upwards from the top A of a tower. It reaches the ground in t_1 seconds. If it is thrown vertically downwards from A with the same speed it reaches the ground in t_2, seconds. If it is allowed to fall freely from A, then the time it takes to reach the ground is given by :

(1) $t = \frac{t_1 + t_2}{2}$ (2) $t = \frac{t_1 - t_2}{2}$

(3) $t = \sqrt{t_1 t_2}$ (4) $t = \sqrt{\frac{t_1}{t_2}}$

PMP307

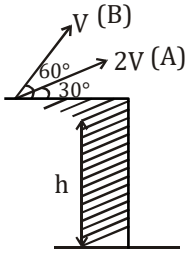
7. The ceiling of a hall is 40m high. For maximum horizontal distance, the angle at which the ball can be thrown with a speed of 56 m/s without hitting the ceiling of the hall is (take $g = 9.8\ m/s^2$) :

(1) 25° (2) 30°

(3) 45° (4) 60°

PMP308

8. Two particles A & B are projected from a building. A is projected with speed 2V making an angle 30° with horizontal & B with speed V making an angle 60° with horizontal. Which particle will hit the ground earlier



(1) Particle A

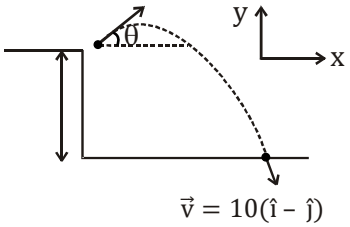
(2) Particle B

(3) Particle A & B will hit at same time.

(4) None of these

PMP309

9. A ball was thrown from height H and the ball hit the floor with velocity $10(\hat{i}-\hat{j})$ after 1.5 sec of its projection. Find initial speed of ball.



(1) $10\sqrt{5}$ m/s (2) $5\sqrt{5}$ m/s
(3) 15 m/s (4) 30 m/s

PMP310

10. In the above question, the angle of projection θ is:

(1) $\theta = \tan^{-1}\left(\frac{3}{2}\right)$ (2) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$
(3) $\theta = \tan^{-1}(2)$ (4) $\theta = \tan^{-1}(1)$

PMP311

11. In the above question 9 the height H is :

(1) 3.75 m (2) 2.25 m
(3) 1.50 m (4) 4.75 m

PMP312

12. A long horizontal belt is moving from left to right with a uniform speed of 2 m/s. There are two ink marks A and B on the belt 60 m apart. An insect runs on the belt to and from between A and B such that its speed relative to the belt is constant and equals 4 m/s. When the insect is moving on the belt in the direction of motion of the belt, its speed as observed by a person standing on ground will be

(1) 6 m/s (2) 2 m/s
(3) 1.5 m/s (4) 4 m/s

PMP313

13. In above question 12 if A lies to the left of B, then :

(1) time taken by insect to travel from A to B and that taken to travel from B to A are equal
(2) time taken by insect to travel from A to B is less than that taken to travel from B to A
(3) time taken by insect to travel from A to B is more than that taken to travel from B to A
(4) none of the above

PMP314

14. In above question 12 if A lies to the left of B, time taken by the insect to travel from B to A will be :

(1) 12 sec
(2) 15 sec
(3) 18 sec
(4) 21 sec

PMP315

15. A particle travels 10m in first 5 seconds and 10 m in next 3 seconds. Assuming constant acceleration, what is the distance travelled in next 2 second?

(1) 8.3 m
(2) 9.3 m
(3) 10.3 m
(4) None of these

PMP316

16. A balloon rises from the ground with an acceleration of 1.25 m/s^2. After 8s, a stone is released from the balloon. The stone will – (taking g = 10 m/s^2)

(1) have a displacement of 50 m
(2) cover a distance of 40 m in reaching the ground
(3) reach the ground in 4 s
(4) begin to move downwards after being released

PMP317

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	3	3	3	2	2	2	2	1	1	1	2	1
Question	16														
Answer	3														

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Exercise - I Units, Dimensions and Measurement

1. Which of the following does not have the same unit as others?
(1) watt-s (2) kilowatt-hour
(3) eV (4) J-s
PPM201

2. In a particular system the units of length mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to
(1) 0.1 N (2) 1 N
(3) 10 N (4) 100 N
PPM202

3. Suppose refractive index μ is given as $\mu = A + B/\lambda^2$, where A and B are constants and λ is wavelength then the dimensions of B are same as that of **:-**
(1) wavelength (2) pressure
(3) area (4) volume
PPM203

4. The units of three physical quantities x, y and z are gcm^2s^{-5}, gs^{-1} and cms^{-2} respectively. The relation among the units of x, y and z is
(1) $z = x^2y$ (2) $y^2 = xz$
(3) $x = yz^2$ (3) $x = y^2z$
PPM204

5. The period of oscillation of a simple pendulum in an experiment is recorded as 2.63s, 2.56s, 2.42s, 2.71s and 2.80s respectively. The average absolute error is
(1) 0.1s (2) 0.11s
(3) 0.01s (4) 1.0s
PPM205

6. Time period of oscillation of a simple pendulum is found to be 2.62 second in an experiment and the mean error calculated is 0.11s. Which one of following is correctly representing result of the experiment?
(1) (2.6 ± 0.11)s (2) (2.6 ± 0.1)s
(3) (2.60 ± 0.11)s (4) (2.62 ± 0.1)s
PPM206

7. The length, breadth and thickness of a strip are
(10.0 ± 0.1) cm
(1.00 ± 0.01) cm and
(0.100 ± 0.001) cm
respectively. The most probable error in its volume will be
(1) ± 0.03 cm^3 (2) ± 0.111 cm^3
(3) ± 0.012 cm^3 (4) None of these
PPM207

8. A wire has a mass (0.3 ± 0.003) g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of its density is–
(1) 1 (2) 2 (3) 3 (4) 4
PPM208

9. In a vernier callipers, one main scale division is x cm and n divisions of the vernier scale coincide with (n–1) divisions of the main scale. The least count (in cm) of the callipers is :-
(1) $\left(\frac{n-1}{n}\right)x$ (2) $\frac{nx}{(n-1)}$
(3) $\frac{x}{n}$ (4) $\frac{x}{(n-1)}$
PPM209

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	4	1	3	3	2	2	1	4	3

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Exercise - II Units, Dimensions and Measurement

1. The dimensional formula for Planck's constant h and gravitational constant G respectively are :-
(1) $[ML^3T^{-2}]$, $[M^{-1}L^2T^{-3}]$
(2) $[ML^2T^{-1}]$, $[M^{-1}L^3T^{-2}]$
(3) $[ML^3T^{-2}]$, $[M^{-1}L^2T^2]$
(4) $[MLT^{-3}]$, $[M^{-1}L^3T^{-3}]$

PPM210

2. If energy (E), velocity (V) and time (T) were chosen as fundamental physical quantities for measurement, then the dimensional formula for mass will be : -
(1) $[E^1 V^2 T^1]$ (2) $[E^2 V^{-2} T^0]$
(3) $[E^1 V^{-2} T^0]$ (4) $[E^{-1} V^2 T^1]$

PPM211

3. The force of interaction between two atoms is given by $F = \alpha\beta \exp\left(\frac{-x^2}{\alpha kT}\right)$. Where x is the distance, k is the Boltzmann constant and T is temperature and α and β as two constants. The dimensional formula of β is -
(1) $[M^2L^2T^{-2}]$ (2) $[M^2LT^{-4}]$
(3) $[M^0L^2T^{-4}]$ (4) $[MLT^{-2}]$

PPM212

4. The length of a cylinder is measured with a metre rod having least count 0.1 cm. Its diameter is measured with vernier callipers having least count 0.01 cm. Given the length is 5.0 cm. and radius is 2.00 cm. The percentage error in the calculated value of volume will be –
(1) 2% (2) 1% (3) 3% (4) 4%

PPM213

5. Given below are two statements :
Statement (I) : Planck's constant and angular momentum have same dimensions.
Statement (II) : Linear momentum and moment of force have same dimensions.
In the light of the above statements, choose the correct answer from the options given below :
(1) Statement I is true but Statement II is false
(2) Both Statement I and Statement II are false
(3) Both Statement I and Statement II are true
(4) Statement I is false but Statement II is true

PPM214

6. Given below are two statements : one is labelled as Assertion(A) and the other is labelled as Reason (R).
Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.
Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

PPM215

7. Consider two physical quantities A and B related to each other as $E = \frac{B - x^2}{At}$ where E, x and t have dimensions of energy, length and time respectively. The dimension of AB is
(1) $[L^{-2}M^1T^0]$ (2) $[L^2M^{-1}T^1]$
(3) $[L^{-2}M^{-1}T^1]$ (4) $[L^0M^{-1}T^1]$

PPM216

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	2	3	2	3	1	2	2

Exercise - I — Laws of Motion and Friction

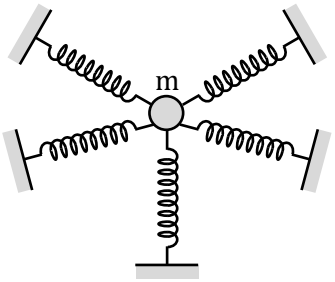
1. Three forces P, Q & R are acting at a point in a plane. The angle between P and Q, Q and R are 150° and 120° respectively. Then for equilibrium, forces P, Q & R are in the ratio :

(1) 1 : 2 : 3 (2) 1 : 2 : $\sqrt{3}$

(3) 3 : 2 : 1 (4) $3^{1/2}$: 2 : 1

PLM295

2. A sphere of mass m is kept in equilibrium with the help of several springs as shown in the figure. Measurements show that one of the springs applies a force $\vec{F}$ on the sphere. With what acceleration the sphere will move immediately after this particular spring is cut?



(1) zero

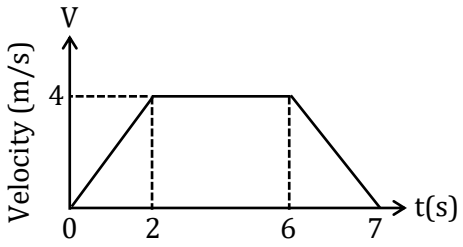
(2) $\vec{F}/m$

(3) $-\vec{F}/m$

(4) insufficient information

PLM296

3. A lift of mass 100 kg starts moving from rests in the upward direction. Fig shows the variation if speed of the lift, T_1, T_2 and T_3 stand for tension in the rope form zero to two seconds, two to six seconds, six to seven seconds respectively then :-



(1) $T_1 : T_2 : T_3 :: 1 : 1 : 1$

(2) $T_1 : T_2 : T_3 :: 6 : 5 : 3$

(3) $T_1 : T_2 : T_3 :: 3 : 5 : 6$

(4) $T_1 : T_2 : T_3 :: 6 : 5 : 6$

PLM297

4. A body of mass m collides against a wall with a velocity v and rebounds with the same speed. Magnitude of change in its momentum is

(1) 2 mv (2) mv

(3) –mv (4) Zero

PLM298

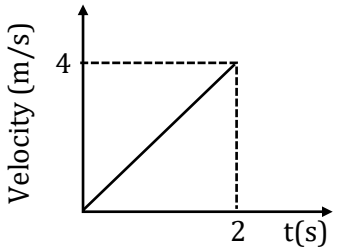
5. When a constant force is applied to a body, it moves with uniform: -

(1) Acceleration (2) Velocity

(3) Speed (4) Momentum

PLM299

6. For a body of 25 kg mass, the velocity-time graph is shown in figure. The force acting on the body is: -



(1) 25 N (2) 50 N

(3) 12.5 N (4) 100 N

PLM300

7. A satellite in force free space sweeps stationary interplanetary dust at a rate $(dM/dt) = +\alpha v^2$. Here v is the velocity. The acceleration of satellite of mass M is: -

(1) $-2\alpha v^2 / M$ (2) $-3\alpha v^2 / M$

(3) $-\alpha v^3 / M$ (4) $-\alpha v^2$

PLM301

8. A bullet of mass 10 gm is fired from a gun of mass 1kg. If the recoil velocity is 5 m/s, the muzzle velocity is

(1) 0.05 m/s (2) 5 m/s

(3) 50 m/s (4) 500 m/s

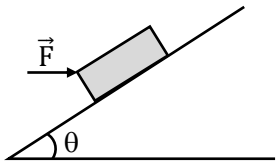
PLM302

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9. Action and reaction: - (For a given system)
(a) Act on the same object
(b) Have same direction
(c) Have different magnitudes
(d) Have non-zero resultant
(1) a, b, c
(2) b, c, d
(3) All of the above
(4) None of the above

PLM303

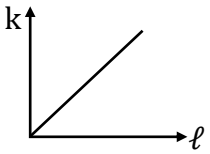
10. The figure shows a horizontal force $\vec{F}$ acting on a block of mass M on an inclined plane (angle θ). What is the normal reaction on the block?

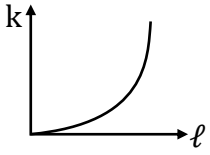


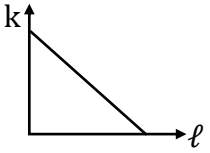
(1) $mg\sin\theta + F\cos\theta$
(2) $mg\sin\theta - F\cos\theta$
(3) $mg\cos\theta - F\sin\theta$
(4) $mg\cos\theta + F\sin\theta$

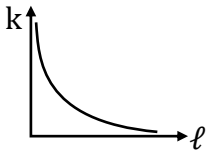
PLM304

11. Which of the following graph depicts spring constant k versus length ℓ of the spring correctly?

(1) 

(2) 

(3) 

(4) 

PLM305

12. The force exerted by a person on the floor of an elevator is less than the weight of the person if the elevator is :-
(a) Going up and slowing down
(b) Going up and speeding up
(c) Going down and slowing down
(d) Going down and speeding up
(1) a, c (2) b, c (3) a, d (4) b, d

PLM306

13. A block of mass 4 kg is placed on the floor. The coefficient of static friction is 0.4. Force of 8 N is applied on the block. The force of friction between the block and the floor is
(1) 16 N (2) 8.0 N
(3) 2.0 N (4) zero

PLM307

Exercise-I — ANSWER KEY

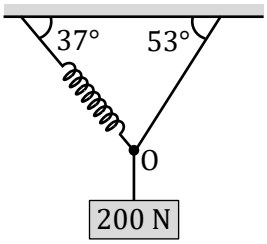
Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	4	3	2	1	1	2	3	4	4	4	4	3	2

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Exercise - II

Laws of Motion and Friction

1. In the set-up shown, a 200 N block is supported in equilibrium with the help of strings and a spring. all knotted at point O. Extension in the spring is 4 cm. Force constant of the spring is closest to [g = 10 m/s²]



(1) 30 N/m (2) 2500 N/m
(3) 3000 N/m (4) 4000 N/m

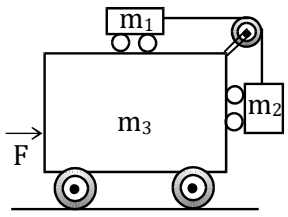
PLM308

2. A pulley is attached to the ceiling of a lift moving upwards. Two particles are attached to the two ends of a string passing over the pulley. The masses of the particles are in the ratio 2 : 1. If the acceleration of the particles relative to the lift is g/2, then the acceleration of the lift will be

(1) g (2) $\frac{g}{2}$ (3) $\frac{g}{3}$ (4) $\frac{g}{4}$

PLM309

3. All surfaces are assumed to be frictionless. Calculate the horizontal force F that must be applied so that m_1 and m_2 do not move relative to m_3 is :–



(1) $(m_1 + m_2 + m_3)\frac{m_2 g}{m_1}$

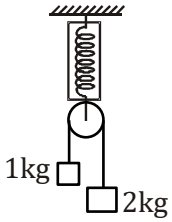
(2) $(m_1 + m_2)\frac{m_2 g}{m_1}$

(3) $(m_2 + m_3)\frac{m_1 g}{m_2}$

(4) $(m_1 + m_3)\frac{m_1 g}{m_2}$

PLM310

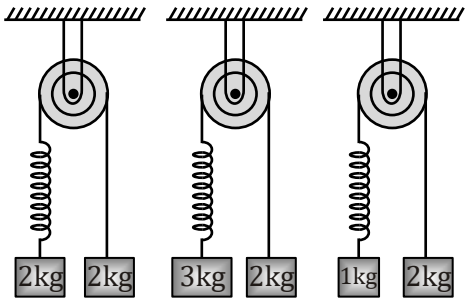
4. Two masses of 1 kg and 2 kg are attached to the ends of a massless string passing over a pulley of negligible weight. The pulley itself is attached to a light spring balance as shown in figure. The masses start moving; during this interval the reading of spring balance will be:-



(1) More than 3 kg.
(2) Less than 3 kg.
(3) Equal to 3 kg.
(4) None of the above

PLM311

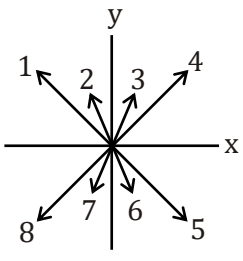
5. Same spring is attached with 2 kg, 3 kg and 1 kg blocks in three different cases as shown in figure. If x_1, x_2 and x_3 be the respective extensions in the spring in these three cases, then :-



(1) $x_1 = 0, x_3 > x_2$ (2) $x_2 > x_1 > x_3$
(3) $x_3 > x_1 > x_2$ (4) $x_1 > x_2 > x_3$

PLM312

6. Two forces $\vec{F}_1 = (3N)\hat{i} - (4N)\hat{j}$ and $\vec{F}_2 = -(1N)\hat{i} - (2N)\hat{j}$ act on a point object. In the given figure which of the eight vectors represents $\vec{F}_1$ and $\vec{F}_2$? What is the magnitude of the net forces?



(1) 4 and 8; $\sqrt{19}$ N (2) 3 and 6;5 N
(3) 5 and 7 ; $\sqrt{40}$ N (4) 2 and 6 ; 7 N

PLM313

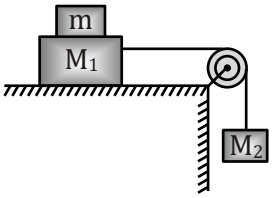
7. Two weights w_1 and w_2 are connected by a light thread which passes over a light smooth pulley. If the pulley is raised upwards with an acceleration equal to g, then the tension in the thread will be:-

(1) $\frac{2w_1w_2}{w_1+w_2}$ (2) $\frac{w_1w_2}{w_1+w_2}$

(3) $\frac{4w_1w_2}{w_1+w_2}$ (4) $\frac{4w_1w_2}{w_1-w_2}$

PLM314

8. Two blocks of masses M_1 = 4 kg and M_2 = 6 kg are connected by a string of negligible mass passing over a frictionless pulley as shown in the figure below. The coefficient of friction between the block M_1 and the horizontal surface is 0.4. When the system is released, the masses M_1 and M_2 start accelerating. What additional mass m should be placed over M_1 so that the masses (M_1 + m) slide with a uniform speed?

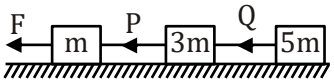


(1) 12 kg (2) 11 kg

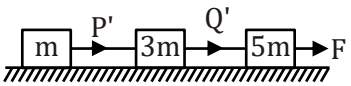
(3) 10 kg (4) 9 kg

PLM315

9. Three blocks of masses m, 3m and 5m are connected by massless strings and pulled by a force F on a frictionless surface as shown in the figure below. The tension P in the first string is 16N.



If the point of application of F is changed as given below, the values of P' and Q' shall be :-

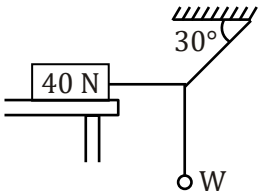


(1) 16N, 10N (2) 10N, 16N

(3) 8 N, 2N (4) None of these

PLM316

10. In the figure given, the system is in equilibrium. What is the maximum value W can have if the friction force on the 40 N block cannot exceed 12.0 N?

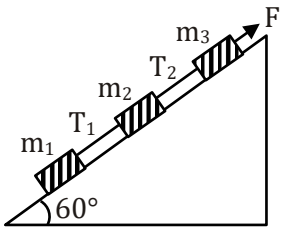


(1) 3.45 N (2) 6.92 N

(3) 10.35 N (4) 12.32 N

PLM317

11. Three blocks, of masses m_1 = 2.0, m_2 = 4.0 and m_3 = 6.0 kg are connected by strings on a frictionless inclined plane of 60^0, as shown in the figure. A force F = 120 N is applied upwards along the incline to the uppermost block, causing an upward movement of the blocks. The connecting cords are light. The values of tensions T_1 and T_2 in the cords are



(1) T_1 = 20 N, T_2 = 60 N

(2) T_1 = 60 N, T_2 = 60 N

(3) T_1 = 30 N, T_2 = 50 N

(4) T_1 = 20 N, T_2 = 1 00 N

PLM318

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	3	2	1	2	2	3	3	2	4	2	1

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Exercise - I — Work, Energy and Power

1. A body of mass 10 kg is displaced from point A(2, 1, 3) to point B(3, 3, 4) under the effect of a force of magnitude 20 N in the direction of $6\hat{i}+8\hat{j}$. Calculate W.D. by the force :-

(1) 22 J (2) $20\sqrt{6}$ J (3) 44 J (4) Zero

PWP171

2. Work done in time t on a body of mass m which is accelerated from rest to a speed v in time t_1 as a function of time t is given by :-

(1) $\frac{1}{2}m\frac{v}{t_1}t^2$ (2) $m\frac{v}{t_1}t^2$

(3) $\frac{1}{2}\left(\frac{mv}{t_1}t\right)^2 t^2$ (4) $\frac{1}{2}m\frac{v_2}{t_1^2}t^2$

PWP172

3. A particle moves on a rough horizontal ground with some initial velocity say v_0. If $\frac{3}{4}$ of its kinetic energy is lost due to friction in time t_0 then coefficient of friction between the particle and the ground is :-

(1) $\frac{v_0}{2gt_0}$ (2) $\frac{v_0}{4gt_0}$ (3) $\frac{3v_0}{4gt_0}$ (4) $\frac{v_0}{gt_0}$

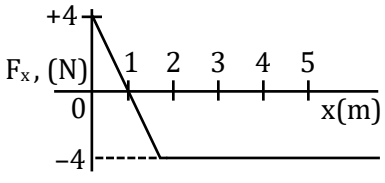
PWP173

4. A body of mass m accelerates uniformly from rest to v_1 in time t_1. The instantaneous power delivered to the body as a function of time t is-

(1) $\frac{mv_1t}{t_1}$ (2) $\frac{mv_1^2t}{t_1^2}$ (3) $\frac{mv_1t^2}{t_1}$ (4) $\frac{mv_1^2t}{t_1}$

PWP174

5. The only force F_x acting on a 2.0 kg body as it moves along the x-axis varies as shown in the figure. The velocity of the body along positive x-axis at x = 0 is 4 m/s. The kinetic energy of the body at x = 3.0 m is :-



(1) 4 J (2) 8 J (3) 12 J (4) 16 J

PWP175

6. A machine which is 75% efficient, uses 12 J of energy in lifting 1 kg mass through a certain distance. The mass is then allowed to fall through the same distance, the velocity at the end of its fall is :-

(1) $\sqrt{12}$ m/s (2) $\sqrt{18}$ m/s

(3) $\sqrt{24}$ m/s (4) $\sqrt{32}$ m/s

PWP176

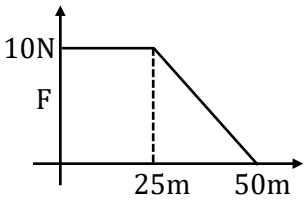
7. A variable force, given by the 2-dimensional vector $\vec{F}=(3x^2\hat{i}+4\hat{j})$, acts on a particle. The force is in newton and x is in metre. What is the change in the kinetic energy of the particle as it moves from the point with coordinates (2,3) to (3,0)? (The coordinates are in meters)

(1) –7 J (2) Zero

(3) +7J (4) +19 J

PWP177

8. A body of mass 5 kg is acted upon by a variable force. The force varies with distance covered by the body. What is the speed of the body when the body has covered 25m assuming that the body starts form rest :-



(1) 10 m/s (2) 20 m/s

(3) 30 m/s (4) 40 m/s

PWP178

9. A retarding force F = 2x + 3x² N acts on a particle in x-direction. Find the work done by this force during a displacement from x = 1m to x = 2.0 m :–

(1) 10 J (2) –10 J (3) 20 J (4) –20 J

PWP179

10. The power input of the pump which is 50% efficient and pumps 200 kg water to a height 20 m in 10 sec : -

(1) 4 KW (2) 6 KW (3) 7 KW (4) 8 KW

PWP180

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	3	4	1	2	3	2	3	1	2	4

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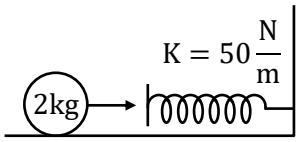
Exercise - II — Work, Energy and Power

1. If momentum is increased $(2)^{1/4}$ times the percentage change in kinetic energy is :
(1) 41.4 % (2) 100%
(3) 14104% (4) 82.6%
PWP181

2. The potential energy of a certain spring when stretched through a distance s is 10 Joule. The amount of work done (in Joule) that must be done on this spring to stretch it through additional distance s will be: -
(1) 30 (2) 40 (3) 10 (4) 20
PWP182

3. The potential energy of a particle varies with x according to the relation $U(x) = x^2 - 4x$. The point x = 2 is a point of : -
(1) stable equilibrium
(2) unstable equilibrium
(3) neutral equilibrium
(4) none of above
PWP183

4. A mass of 2 kg moving with a speed of 10m/sec on a horizontal smooth surface compresses a spring of spring constant $K = 50\frac{N}{m}$. The maximum compression of spring: -



(1) 2m (2) 1.5m
(3) 14m (4) 6m
PWP184

5. Two bodies of mass 4 g and 25 g are moving with equal kinetic energies. The ratio of magnitude of their linear momentum is :
(1) 3 : 5 (2) 5 : 4
(3) 2 : 5 (4) 4 : 5
PWP185

6. The potential energy function (in J) of a particle in a region of space is given as $U = (2x^2 + 3y^3 + 2z)$. Here x, y and z are in meter. The magnitude of x - component of force (in N) acting on the particle at point P (1, 2, 3) m is :
(1) 2 (2) 6
(3) 4 (4) 8
PWP186

7. An artillery piece of mass M_1 fires a shell of mass M_2 horizontally. Instantaneously after the firing, the ratio of kinetic energy of the artillery and that of the shell is :
(1) $\frac{M_1}{(M_1+M_2)}$ (2) $\frac{M_2}{M_1}$
(3) $\frac{M_2}{(M_1+M_2)}$ (4) $\frac{M_1}{M_2}$
PWP187

8. A body of mass 2 kg begins to move under the action of a time dependent force given by $\vec{F} = \left(6t\,\hat{i} + 6t^2\,\hat{j}\right)N$. The power developed by the force at the time t is given by:
(1) $\left(6t^4 + 9t^5\right)W$ (2) $\left(3t^3 + 6t^5\right)W$
(3) $\left(9t^5 + 6t^3\right)W$ (4) $\left(9t^3 + 6t^5\right)W$
PWP188

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	1	1	1	1	3	3	2	4

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Exercise - I — Circular Motion

1. A wheel has a constant angular acceleration of 3.0 rad/s², During a certain 4.0 s interval, it turns through an angle of 120 rad. Assuming that at t = 0, angular speed ω_0 = 3 rad/s, how long, is motion before the start of this 4.0 second interval?

(1) 7 sec. (2) 9 sec.
(3) 4 sec. (4) 10 sec.

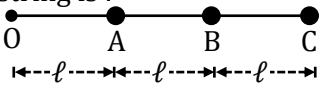
PCM129

2. Keeping the banking angle of the road constant, the maximum safe speed of passing vehicles is to be increased by 10%. The radius of curvature of the road will have to be changed from 20 m to :-

(1) 16 m (2) 18 m
(3) 24.20 m (4) 30.5 m

PCM130

3. Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a horizontal plane. If the velocity of the outermost particle is v_0, then the ratio of tensions in the three sections of the string is :-



(1) 3 : 5 : 7 (2) 3 : 4 : 5
(3) 7 : 11 : 6 (4) 3 : 5 : 6

PCM131

4. A mass m is attached to the end of a rod of length ℓ. The mass goes around a verticle circular path with the other end hinged at the centre. What should be the minimum velocity of mass at the bottom of the circle so that the mass completes the circle?

(1) $\sqrt{4g\ell}$ (2) $\sqrt{3g\ell}$
(3) $\sqrt{5g\ell}$ (4) $\sqrt{g\ell}$

PCM132

5. A stone is tied to a string of length 'ℓ' and is whirled in a vertical circle with the other end of the string as the centre. At a certain instant of time, the stone is at its lowest position and has a speed 'u'. The magnitude of the change in velocity as it reaches a position where the string is horizontal (g being acceleration due to gravity) is :–

(1) $\sqrt{u^2 - g\ell}$ (2) $u - \sqrt{u^2 - 2g\ell}$
(3) $\sqrt{2g\ell}$ (4) $\sqrt{2(u^2 - g\ell)}$

PCM133

Exercise-I — ANSWER KEY

Question	1	2	3	4	5
Answer	1	3	4	1	4

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Exercise - II — Circular Motion

1. Angular velocity of hour hand of a clock is :-

(1) $\frac{\pi}{30}$ rad / sec

(2) 8π rad/sec

(3) $\frac{2\pi}{1800}$ rad / sec

(4) None of the above

PCM134

2. A body crosses the topmost point of a vertical circle with critical speed. Its centripetal acceleration, when the string is horizontal will be :-

(1) 6g (2) 3g (3) 2g (4) g

PCM135

3. A weightless thread can support tension up to 30 N. A particle of mass 0.5 kg is tied to it and is revolved in a circle of radius 2 m in a vertical plane. If g = 10 m/s², then the maximum angular velocity of the stone will be: –

(1) 5 rad/s (2) $\sqrt{30}$ rad / s

(3) $\sqrt{60}$ rad / s (4) 10 rad/s

PCM136

4. If angular velocity of a particle depends on the angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration at θ = 1 rad is : -

(1) 8 rad/s²

(2) 10 rad/s²

(3) 12 rad/s²

(4) None of the above

PCM137

5. A stone is rotated in a vertical circle. Speed at bottom most point is $\sqrt{8gR}$, where R is the radius of circle. The ratio of tension at the top and the bottom is

(1) 1 : 2

(2) 1 : 3

(3) 2 : 3

(4) 1 : 4

PCM138

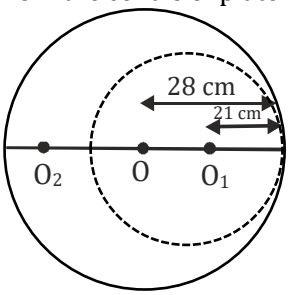
Exercise-II — ANSWER KEY

Question	1	2	3	4	5
Answer	4	2	1	3	2

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Exercise - I — Collisions and Centre of Mass

1. A circular plate of uniform thickness has a diameter 56 cm. A circular portion of diameter 42 cm is removed from one edge as shown in the figure. The centre of mass of the remaining portion from the centre of plate will be :



(1) 5 cm (2) 7 cm
(3) 9 cm (4) 11 cm

PSC136

2. A shell is fired from a canon with velocity V m/s at an angle θ with the horizontal direction. At the highest point in its path with same speed it explodes into two pieces of equal masses. One of the pieces retraces its path to the cannon. The speed in m/s of the other piece immediately after the explosion is

(1) $\left(\frac{\sqrt{3}}{2}\right)V\cos\theta$ (2) $3V\cos\theta$

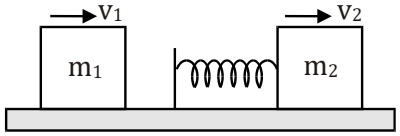
(3) $2V\cos\theta$ (4) $\left(\frac{3}{2}\right)V\cos\theta$

PSC137

3. After falling from a height of 5m a ball strikes the roof of a lift. If at the time of collision, lift is moving in the upward direction with a velocity of 1m/s, then the velocity with which the ball rebounds after collision will be – (e = 1)
(1) 11 m/s downwards
(2) 12 m/s upwards
(3) 13 m/s upwards
(4)12 m/s downwards

PSC138

4. Two masses m_1 = 2kg and m_2 = 5kg are moving on a frictionless surface with velocities 10m/s and 3 m/s respectively. m_2 is ahead of m_1. An ideal spring of spring constant k = 1120 N/m is attached on the back side of m_2. The maximum compression of the spring will be :-



(1) 0·51 m (2) 0·062 m
(3) 0·25 m (4) 0·72 m

PSC139

5. A frictionless steel ball of radius 2cm, moving on a horizontal plane with a velocity of 5cm/s, collides head–on with another stationary steel ball of radius 3cm. The velocities of two bodies after collision will respectively be (in cm/s) (e = 1)
(1) 2·7, 2·3 (2) –2·7, 2·3
(3) 2·7, –2·3 (4) –2·7, –2·3

PSC140

6. A nucleus of mass number A originally at rest, emits alpha particle with speed v. The recoil speed of the daughter nucleus is :-

(1) $\frac{4v}{A-4}$ (2) $\frac{4v}{A+4}$

(3) $\frac{v}{A-4}$ (4) $\frac{v}{A+4}$

PSC141

7. A sphere of diameter r is cut from a solid sphere of radius r such that the centre of mass of remaining part be at maximum distance from original centre, then this distance is :

(1) $\frac{r}{2}$ (2) $\frac{r}{3}$

(3) $\frac{r}{14}$ (4) none of these

PSC142

8. In the HCl molecule, the separation between the nuclei of the two atoms is about 1.27 Å (1Å = 10^{-10} m). The approximate location of the centre of mass of the molecule from hydrogen atom, assuming the chlorine atom to be about 35.5 times massive as hydrogen is
(1) 1 Å (2) 2.5 Å (3) 1.24 Å (4) 1.5 Å

PSC143

9. A man m = 80 kg is standing on a trolley of mass 320 kg on a smooth surface. If man starts walking on trolley along rails at a speed of 1 m/s, then after 4 sec, his displacement relative to ground is :-

(1) 4 m (2) 4.8 m (3) 3.2 m (4) 6 m

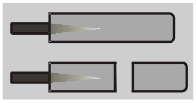
PSC144

10. A neutron makes a head on elastic collision with a stationary deuteron. The fractional energy loss of the neutron in the collision is: -

(1) 16/82 (2) 8/9 (3) 8/27 (4) 2/3

PSC145

11. A cricket bat is cut at the location of its centre of mass as shown. Then :-



(1) The two pieces will have the same mass
(2) The bottom piece will have larger mass
(3) The handle piece will have larger mass
(4) Mass of handle piece is double the mass of bottom piece

PSC146

12. If linear density of a rod of length 3 m varies as $\lambda = 2 + x$, then the position of the centre of gravity of the rod is :

(1) $\frac{7}{3}$m (2) $\frac{12}{7}$m (3) $\frac{10}{7}$m (4) $\frac{9}{7}$m

PSC147

13. A body of mass 2.0 kg makes an elastic collision with another body at rest and continues to move in the original direction but with one-fourth of its original speed v. What is the mass of other body and the speed of the center of mass of two bodies?

(1) 1.0 kg and $\frac{2}{3}v$ (2) 1.2 kg and $\frac{5}{8}v$

(3) 1.4 kg and $\frac{10}{17}v$ (4) 1.5 kg and $\frac{4}{7}v$

PSC148

14. A uniform rod of length 1.0 metre is bent at its midpoint to make 90° angle. The distance of the centre of mass from the centre of the rod is : -

(1) 35.3 cm (2) 25.2 cm (3) 17.7 cm (4) zero

PSC149

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer	3	2	2	3	2	1	3	3	3	2	2	2	2	3

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Exercise - II

Collisions and Centre of Mass

1. Two particles moving in the same direction with speeds 4 m/s and 2 m/s collide elastically (the collision being head on). After collision, the velocity of first particle becomes 3 m/s in the same direction. The velocity of the second sound be : -
(1) 2 m/s in same direction
(2) 4 m/s in same direction
(3) 4 m/s in opposite direction
(4) 5 m/s in same direction

PSC150

2. Two particles of masses 2 kg and 4 kg are approaching towards each other with accelerations of 1 m/s^2 and 2 m/s^2 respectively, on a smooth horizontal surface. Find the magnitude of acceleration of centre of mass of the system.
(1) 1 (2) 2 (3) 3 (4) 4

PSC151

3. A system consists of 3 particles each of mass m and located at (1, 1) (2, 2) (3, 3). The co-ordinates of the centre of mass are :
(1) (6, 6) (2) (3, 3)
(3) (2, 2) (4) (1, 1)

PSC152

4. A ball is dropped from a height of 1 m. If coefficient of restitution between the surface and the ball is 0.6, the ball rebounds to a height of : -
(1) 0.6 m (2) 0.4 m
(3) 1 m (4) 0.36 m

PSC153

5. A body of mass 1000 kg is moving horizontally with a velocity 6 m/s. If 200 kg extra mass is added, the final velocity (in m/s) is:
(1) 6 (2) 2 (3) 3 (4) 5

PSC154

6. A simple pendulum of length 1 m has a wooden bob of mass 1 kg. It is struck by a bullet of mass 10^{-2} kg moving with a speed of 2×10^2 ms^{-1}. The bullet gets embedded into the bob. The height to which the bob rises before swinging back is. (use g = 10 m/s^2)
(1) 0.30 m (2) 0.20 m
(3) 0.35 m (4) 0.40 m

PSC155

7. The identical spheres each of mass 2M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 4 m each. Taking point of intersection of these two sides as origin, the magnitude of position vector of the centre of mass of the system is $\frac{4\sqrt{2}}{x}$, where the value of x is _____
(1) 8 (2) 6 (3) 4 (4) 3

PSC156

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	4	1	3	4	4	2	4

Exercise - I — Rotational Motion

1. A ladder rests against a frictionless vertical wall, with its upper end 6m above the ground and the lower end 4m away from the wall. The weight of the ladder is 500 N and its centre of gravity at (1/3)rd distance from the lower end. Wall's reaction will be, in N (approx.) :-

(1) 111 (2) 333 (3) 222 (4) 129

PRM247

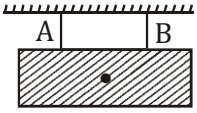
2. A string is wound round the rim of a mounted fly wheel disc of mass 20 kg and radius 20 cm. A steady pull of 25 N is applied on the cord. Neglecting friction and mass of the string, the angular acceleration of the wheel is :-

(1) 50 rad/s^2 (2) 25 rad/s^2

(3) 12.5 rad/s^2 (4) 6.25 rad/s^2

PRM248

3. The figure shows a horizontal block of mass M suspended by two wires A and B. The centre of mass of the block is closer to B than A. (i) Is the magnitude of the torque due to wire A is greater, less or equal to that due to B w.r.t. centre of mass? (ii) Which wire A or B exerts more force on the block?

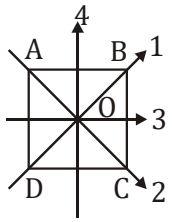


(1) (i) greater (ii) B (2) (i) equal (ii) B

(3) (i) less (ii) A (4) (i) greater (ii) A

PRM249

4. The moment of inertia of a thin square plate ABCD (shown in figure below) of uniform thickness about an axis passing through the centre O and perpendicular to the plane of the plate is



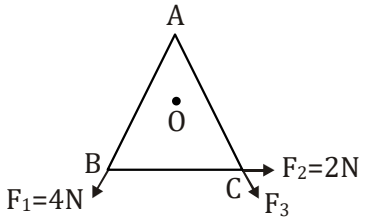
(AB = BC = CD = DA)

(1) $I_1 + I_2$ (2) $I_1 + I_2 + I_3 + I_4$

(3) $I_1 - I_3$ (4) $I_2 - I_4$

PRM250

5. In an equilateral triangle ABC, F_1, F_2 and F_3 are three forces acting along the sides AB, BC and AC as shown in the given figure. What should be the magnitude of F_3 so that the total torque about O is zero :-



(1) 2 N (2) 4 N (3) 6 N (4) 8 N

PRM251

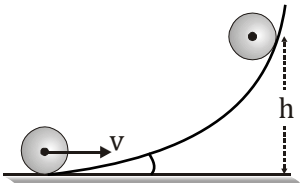
6. A particle of mass m is projected with a velocity v making an angle 45° with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle is at its maximum height h, is :

(1) zero (2) $\frac{mv^3}{4\sqrt{2g}}$

(3) $\frac{mv^3}{\sqrt{2g}}$ (4) $m^2\sqrt{2gh^3}$

PRM252

7. A disc of mass M and radius R rolls on a horizontal surface and then rolls up an inclined plane as shown in the figure. If the velocity of the disc is v, the height to which the disc will rise will be :-



(1) $\frac{3v^2}{2g}$ (2) $\frac{3v^2}{4g}$

(3) $\frac{v^2}{4g}$ (4) $\frac{v^2}{2g}$

PRM253

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	1	3	2	1	3	2	2

Exercise - II Rotational Motion

1. The angular velocity of a body is $\vec{\omega} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and a torque $\vec{\tau} = \hat{i} + 2\hat{j} + 3\hat{k}$ acts on it. The rotational power will be : -

(1) 20 unit (2) 15 unit

(3) $\sqrt{17}$ unit (4) $\sqrt{406}$ unit

PRM254

2. If I_1, I_2 and I_3 are moments of inertia of solid sphere, hollow sphere and a ring of same mass and radius about geometrical axis, which of the following statements hold good?

(1) $I_1 > I_2 > I_3$ (2) $I_3 > I_2 > I_1$

(3) $I_2 > I_1 > I_3$ (4) $I_2 > I_3 > I_1$

PRM255

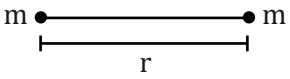
3. The ratio of the radii of gyration of a circular disc to that of a circular ring, each of same mass and radius around their natural axes passing through centre is :-

(1) $\sqrt{2}:1$ (2) $\sqrt{2}:\sqrt{3}$

(3) $\sqrt{3}:\sqrt{2}$ (4) $1:\sqrt{2}$

PRM256

4. Two same masses m are placed at a distance r from each other. Find out moment of inertia of system, about an axis passing through their centre of mass and perpendicular the line joining the two masses.



(1) $\frac{mr^2}{4}$ (2) $2mr^2$ (3) $\frac{mr^2}{2}$ (4) $4mr^2$

PRM257

5. A wire of mass m and length ℓ is bent in the form of a circular ring. The moment of inertia of the ring about its axis is :-

(1) $m\ell^2$

(2) $\frac{m\ell^2}{4\pi^2}$

(3) $\frac{m\ell^2}{2\pi^2}$

(4) $\frac{m\ell^2}{8\pi^2}$

PRM258

6. A circular disc is rolling on a horizontal plane. Its total kinetic energy is 150 J. What is its translational KE?

(1) 200 J

(2) 100 J

(3) 125 J

(4) None of these

PRM259

7. Which of the following is true for pure rolling on ground :-

(1) $V_{cm} = R\omega$

(2) $a_{cm} = R\alpha$

(3) Both (1) and (2)

(4) None of these

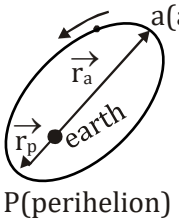
PRM260

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	1	2	4	3	2	2	3

Exercise - I — Gravitation

1. Consider a satellite orbiting the earth as shown in the figure below. Let L_a and L_p represent the angular momentum of the satellite about the earth when at aphelion and perihelion respectively. Consider the following relations.



(i) $\vec{L}_a = \vec{L}_p$ (ii) $\vec{L}_a = -\vec{L}_p$

(iii) $\vec{r}_a \times \vec{L}_a = \vec{r}_p \times \vec{L}_p$

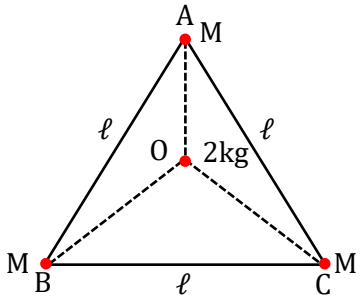
Which of the above relations is/are true?

(1) (i) only (2) (ii) only

(3) (iii) only (4) (i) and (iii)

PGR199

2. Three masses each of mass M are placed at the vertices of an equilateral triangle ABC of side ℓ as shown in figure. The force acting on mass 2 kg. Placed at the centroid of the triangle is



(1) zero (2) $\frac{2Gm^2}{\ell^2}$

(3) $\frac{4Gm^2}{\ell^2}$ (4) $\frac{6Gm^2}{\ell^2}$

PGR200

3. If R is the radius of the Earth and g the acceleration due to gravity on the Earth's surface, the mean density of the Earth is

(1) $4\pi G/3gR$ (2) $4\pi R/4gG$

(3) $3g/4\pi RG$ (4) $\pi RG/12G$

PGR201

4. Imagine a new planet having the same density as that of Earth but its radius is 3 times bigger than the Earth in size. If the acceleration due to gravity on the surface of Earth is g and that on the surface of the new planet is g', then :

(1) g' = 9g (2) g' = g/9

(3) g' = 27g (4) g' = 3g

PGR202

5. Find the percentage decrement in the weight of a body when taken to a height of 16km above the surface of Earth (Radius of Earth is 6400km)

(1) 0.5% (2) 0.25%

(3) 1% (4) 0.75%

PGR203

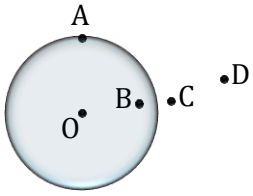
6. What should be angular velocity of Earth so that weight of a man at equator becomes $3/5^{th}$ of its present values. (write your answer in terms of g and R).

(1) $\sqrt{\frac{5}{2}gR}$ (2) $\sqrt{\frac{5}{2}\frac{g}{R}}$

(3) $\sqrt{\frac{2}{5}\frac{g}{R}}$ (4) $\sqrt{5gR}$

PGR204

7. In the given figure there is a hollow sphere then on which point gravitational potential due to sphere will be same.



(1) O, A, B, C (2) O, A, B

(3) O, B (4) A, B

PGR205

8. If velocity given to an object from the surface of the earth is n times the escape velocity then what will be its residual velocity at infinity?

(1) $(\sqrt{n^2-1})V_e$ (2) $(\sqrt{V_e^2-1})n^2$

(3) $(\sqrt{n^2+1})V_e$ (4) $(\sqrt{n^2-1}) \times \frac{1}{V_e}$

PGR206

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9. The escape velocity of a particle of mass m varies as -

(1) m^2 (2) m (3) m^{-1} (4) m^0

PGR207

10. The kinetic energy needed to project a body of mass m from surface of earth (radius R) to infinity is -

(1) $\frac{mgR}{2}$ (2) 2mgR

(3) mgR (4) $\frac{mgR}{4}$

PGR208

11. Two planets are at mean distance d_1 and d_2 from the sun and their frequencies are n_1 and n_2 respectively then.

(1) $n_1^2 d_1^2 = n_2^2 d_2^2$

(2) $n_2^2 d_2^3 = n_1^2 d_1^3$

(3) $n_1 d_1^2 = n_2 d_2^2$

(4) $n_1^2 d_1 = n_2^2 d_2$

PGR209

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	1	1	3	4	1	3	2	1	4	3	2

Exercise - II — Gravitation

1. Masses and radii of earth and moon are M_1, M_2 and R_1, R_2 respectively. The distance between their centre is 'd'. The minimum velocity given to mass 'M' from the mid point of line joining their centre so that it will escape :

(1) $\sqrt{\frac{4G(M_1+M_2)}{d}}$ (2) $\sqrt{\frac{4G}{d}\frac{M_1M_2}{(M_1+M_2)}}$

(3) $\sqrt{\frac{2G}{d}\left(\frac{M_1+M_2}{M_1M_2}\right)}$ (4) $\sqrt{\frac{2G}{d}(M_1+M_2)}$

PGR210

2. A planet revolves around the sun in an elliptical orbit. If v_p and v_a are the velocities of the planet at the perigee and apogee respectively, then the eccentricity of the elliptical orbit is given by :

(1) $\frac{v_p}{v_a}$ (2) $\frac{v_a-v_p}{v_a+v_p}$

(3) $\frac{v_p+v_a}{v_p-v_a}$ (4) $\frac{v_p-v_a}{v_p+v_a}$

PGR211

3. Two solid spherical planets of equal radii R having masses 4M and 9M their centre are separated by a distance 6R. A projectile of mass m is sent from the planet of mass 4 M towards the heavier planet. What is the distance r of the point from the lighter planet where the gravitational force on the projectile is zero?

(1) 1.4 R (2) 1.8 R (3) 1.5 R (4) 2.4 R

PGR212

4. A small planet is revolving around a very massive star in a circular orbit of radius R with a period of revolution T. If the gravitational force between the planet and the star were proportional to $R^{-5/2}$, then T would be proportional to

(1) $R^{3/2}$ (2) $R^{3/5}$ (3) $R^{7/2}$ (4) $R^{7/4}$

PGR213

5. A particle falls on earth :

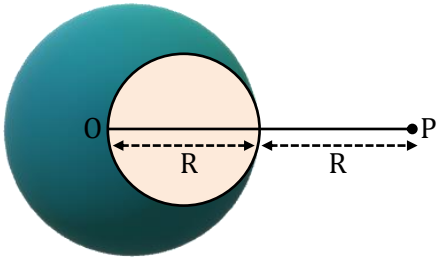
(i) from infinity, (ii) from a height 10 times the radius of earth. The ratio of the velocities gained on reaching at the earth's surface is :

(1) $\sqrt{11}:\sqrt{10}$ (2) $\sqrt{10}:\sqrt{11}$

(3) 10 : 11 (4) 11 : 10

PGR214

6. A solid sphere of uniform density and radius R exerts a gravitational force of attraction F_1 on a particle P, distant 2R from the centre of the sphere. A spherical cavity of radius R/2 is now formed in the sphere as shown in figure. The sphere with cavity now applies a gravitational force F_2 on the same particle P. Find the ratio F_2/F_1.



(1) $\frac{7}{9}$ (2) $\frac{9}{7}$ (3) $\frac{8}{7}$ (4) $\frac{7}{8}$

PGR215

7. In a certain region of space gravitational field is given by I = – (K/r) (Where r is the distance from a fixed point and K is constant). Taking the reference point to be at $r = r_0$ with $V = V_0$. Find the potential at a distance r.

(1) $K\ln\left(\frac{r}{r_0}\right)-V_0$ (2) $K\ln\left(\frac{r}{r_0}\right)+V_0$

(3) $K\ln\left(\frac{r_0}{r}\right)-V_0$ (4) $K\ln\left(\frac{r_0}{r}\right)+V_0$

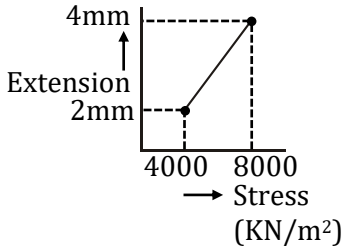
PGR216

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	1	4	4	4	1	1	2

Exercise - I Properties of Matter and Fluid Mechanics

1. In determination of young's modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is '1m'. The curve between extension and stress is depicted then Young's modulus of wire will be : -



(1) 2×10^9 N/m² (2) 1×10^9 N/m²
(3) 2×10^{10} N/m² (4) 1×10^{10} N/m²

PPF256

2. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area 3A. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(1) 6F (2) 9F (3) F (4) 4F

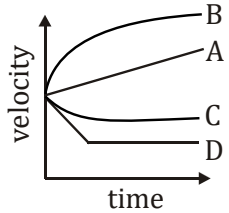
PPF257

3. To what height h should a cylindrical vessel of diameter d be filled with a liquid so that due to liquid force on the vertical surface of the vessel be equal to the force on the bottom :–

(1) h = d (2) h = 2d
(3) h = 3d (4) h = d/2

PPF258

4. A small ball is left in a viscous liquid in a tall container from large height. Correct graph of its velocity with time is :



(1) A (2) B (3) C (4) D

PPF259

5. If a ball of steel (density $\rho = 7.8$ g/cm³) attains a terminal velocity of 10 cm/s when falling in a tank of water (coefficient of viscosity $\eta_{water} = 8.5 \times 10^{-4}$ Pa-s) then its terminal velocity in glycerine ($\rho = 1.2$ g/cm³, $\eta = 13.2$ Pa-s) would be nearly : -

(1) 1.6×10^{-5} cm/s (2) 3.98×10^{-4} cm/s
(3) 6.25×10^{-4} cm/s (4) 1.5×10^{-5} cm/s

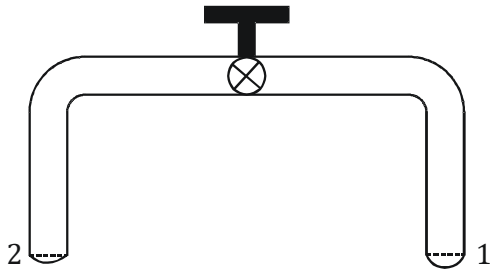
PPF260

6. If more air is pushed in a soap bubble, the pressure in it :

(1) decreases
(2) increase
(3) remains same
(4) becomes zero

PPF261

7. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position.



End 1 has a hemispherical soap bubble of radius r. End 2 has sub–hemispherical soap bubble as shown in figure. Just after opening the valve,

(1) Air from end 1 flows towards end 2. No change in the volume of the soap bubbles.
(2) Air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases.
(3) No change occurs
(4) Air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases.

PPF262

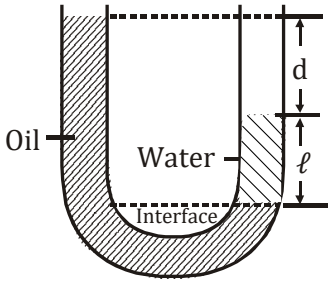
8. A wooden cube just floats inside water when a 200 g mass is placed on it. When the mass is removed the cube is 2 cm above water level. The side of cube is

(1) 5 cm (2) 10 cm

(3) 15 cm (4) 20 cm

PPF263

9. A U-tube contains two liquids in static equilibrium: Water of density ρ_w (= 1000 kg/m^3) is in the right arm, oil of unknown density ρ is the left arm as shown in figure. Measurement give ℓ = 135 mm and d = 12.5 mm. The density of oil is :-



(1) 1092 kg/m^3 (2) 961 kg/m^3

(3) 915 kg/m^3 (4) 843 kg/m^3

PPF264

10. A hole is there in the bottom of the tank having water. If total pressure at bottom is 3 atm (1 atm = 10^5 N/m^2), then velocity of water flowing from hole is

(1) $\sqrt{400}$ m/s (2) $\sqrt{600}$ m/s

(3) $\sqrt{60}$ m/s (4) none of these

PPF265

11. The terminal speed attained by an aluminium sphere of radius 1 mm falling through water at 20°C will be close to (Assume laminar flow, specific gravity of Al = 2.7 and $\eta_{water} = 8 \times 10^{-4}$ Pl.)

(1) 9.2 m/s (2) 6.9 m/s

(3) 4.6 m/s (4) 2.3 m/s

PPF266

12. If the difference between pressure inside and outside of a soap bubble is 6 mm of water and its radius is 8 mm. What is the surface tension in dynes per cm.

(1) 117.6 (2) 256

(3) 378 (4) 450

PPF267

13. Diameter of two limbs of a U-tube are 2 mm and 5 mm, and surface tension is 70 dyne/cm, if water density is 1000 kg/m^2 and g = 10 m/s^2, then water level difference in both the tubes will be :

(1) 8.4 m (2) 8.4 cm

(3) 84 cm (4) 0.84 cm

PPF268

14. A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of water in the capillary tube is 5g. Another capillary tube of radius 2r is immersed in water. The mass of water that will rise in this tube is :-

(1) 2.5 g (2) 5.0 g

(3) 10 g (4) 20 g

PPF269

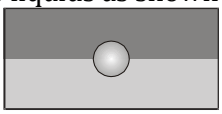
Exercise-I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	1	2	4	3	3	1	2	2	3	1	3	1	4	3	

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Exercise - II Properties of Matter and Fluid Mechanics

1. A solid uniform ball having volume V and density ρ floats at the interface of two immiscible liquids as shown in figure.



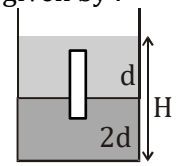
The densities of the upper and the lower liquids are ρ_1 and ρ_2 respectively $\rho_1 < \rho < \rho_2$., such that What fraction of the volume of the ball will be in the lower liquid :–

(1) $\frac{\rho-\rho_2}{\rho_1-\rho_2}$ (2) $\frac{\rho_1}{\rho_1-\rho_2}$

(3) $\frac{\rho_1-\rho}{\rho_1-\rho_2}$ (4) $\frac{\rho_1-\rho_2}{\rho_2}$

PPF270

2. A homogeneous solid cylinder of length L (L < H/2), cross–sectional area A/5 is immersed such that it floats with its axis vertical at the liquid–liquid interface with length L/4 in the denser liquid as shown in the figure. The lower density liquid is open to atmosphere having pressure P_0. Then, density D of solid is given by :–



(1) $\frac{5}{4}$d (2) $\frac{4}{5}$d (3) 4d (4) $\frac{d}{5}$

PPF271

3. A ball rises to surface at a constant velocity in liquid whose density is 4 times than that of the material of the ball. The ratio of the force of friction acting on the rising ball and its weight is:

(1) 3 : 1 (2) 4 : 1 (3) 1 : 3 (4) 1 : 4

PPF272

4. When a large bubble rises from the bottom of a lake to the surface, its radius doubles. If atmospheric pressure is equal to that of column of water height H, then the depth of lake is :

(1) H (2) 2H (3) 7H (4) 8H

PPF273

5. Water stands upto a height h behind the vertical wall of a dam. What is the net horizontal force pushing the dam down by the stream, if width of the dam is σ?
(ρ = density of water)

(1) 2hσg (2) $\frac{h^2\sigma\rho g}{2}$

(3) $\frac{h^2\sigma\rho g}{4}$ (4) $\frac{h\sigma\rho g}{4}$

PPF274

6. A U-tube is partially filled with water. Oil which does not mix with water is next poured into one side, until water rises by 25 cm on the other side. If the density of the oil is 0.8g/cc, the oil level will stand higher than the water level by

(1) 6.25 cm (2) 12.50 cm
(3) 18.75 cm (4) 25.00

PPF275

7. Water is moving with a speed of 5.0 m/s through a pipe will cross sectional area of 4.0 cm^2. The water gradually descends 10 m as the pipe increase in area to 8.0 cm^2. If the pressure at the upper level is 1.5×10^5 Pa, the pressure at lower level will be:-

(1) 2.8×10^5 Pa (2) 2.6×10^5 Pa
(3) 2.4×10^5 Pa (4) 2.1×10^5 Pa

PPF276

8. An air bubble is lying just below the surface of water. The surface tension of water is 70×10^{-3} N/m and atmospheric pressure is 1.013×10^5 N/m^2. If the radius of bubble is 1 mm, then the pressure inside the bubble will be –

(1) 1.0270×10^5 Pa (2) 1.0160×10^5 Pa
(3) 1.0144×10^5 Pa (4) 1.0131×10^5 Pa

PPF277

9. Which one of the following will make its way most easily through the tiny space between the fiber of the clothing –

(1) Glycerine at 20°C (2) Water at 20°C
(3) Soap water at 20°C (4) Water at 100°C

PPF278

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	3	1	1	3	2	2	2	3	4

Exercise - I — Thermal Physics

1. Two rods one of aluminium of length ℓ_1 having coefficient of linear expansion α_a, and other steel of length ℓ_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{\ell_1}{\ell_1+\ell_2}$ is

(1) $\frac{\alpha_s}{\alpha_a+\alpha_s}$ (2) $\frac{\alpha_s}{\alpha_a-\alpha_s}$

(3) $\frac{\alpha_a+\alpha_s}{\alpha_s}$ (4) None of these

PTM507

2. An ideal gas is expanding such that PT^2 = constant. The coefficient of volume expansion of the gas is :

(1) $\frac{1}{T}$ (2) $\frac{2}{T}$ (3) $\frac{3}{T}$ (4) $\frac{4}{T}$

PTM508

3. 50 g of ice at 0° C is mixed with 50 g of water at 100° C. The final temperature of mixture is :–

(1) 0° C

(2) Between 0°C to 20°C

(3) 20°C

(4) Above 20°C

PTM509

4. A sphere, a cube and a thin circular plate all made of same substance and all have same mass. These are heated to 200°C and then placed in a room, then the :

(1) Temperature of sphere drops to room temperature at last.

(2) Temperature of cube drops to room temperature at last

(3) Temperature of thin circular plate drops to room temperature at last

(4) Temperature of all the three drops to room temperature at the same time

PTM510

5. A slab consists of two parallel layers of copper and brass of the same thickness and having thermal conductivities in the ratio 1 : 4. If the free face of brass is at 100°C and that of copper at 0°C, the temperature of interface is -

(1) 80°C (2) 20°C (3) 60°C (4) 40°C

PTM511

6. Out of the metal balls of same diameter one is solid and other is hollow. Both are heated to the same temperature at 300°C and then allowed to cool in the same surroundings then rate of loss of heat will be:

(1) More for hollow sphere

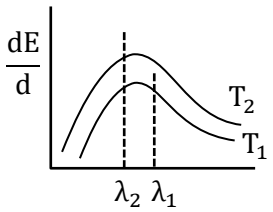
(2) More for solid sphere

(3) Same for both

(4) None of the above

PTM512

7. The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A. At a different temperature T_2, the area is found to be 9A. Then λ_1/λ_2 =



(1) 3 (2) 1/3 (3) $1/\sqrt{3}$ (4) $\sqrt{3}$

PTM513

8. Consider a gas with density ρ and $\bar{c}$ as the root mean square velocity of its molecules contained in a volume. If the system moves as whole with velocity v, then the pressure exerted by the gas is

(1) $\frac{1}{3}\rho\,(\bar{c})^2$ (2) $\frac{1}{3}\rho(\bar{c}+v)^2$

(3) $\frac{1}{3}\rho(\bar{c}-v)^2$ (4) $\frac{1}{3}\rho(c^{-2}-v)^2$

PTM514

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9. Air is filled at 60°C in a vessel of open mouth. The vessel is heated to a temperature T so that 1/4th part of air escapes. The value of T is

(1) 80°C (2) 444°C

(3) 333°C (4) 171°C

PTM515

10. Root mean square velocity for a certain di-atomic gas at room temperature 27°C is found to be 1930 m/sec. The gas is -

(1) H_2 (2) O_2 (3) F_2 (4) Cl_2

PTM516

11. A gas has volume V and pressure P. The total translational kinetic energy of all the molecules of the gas is:–

(1) $\frac{3}{2}$ PV only if the gas is monoatomic.

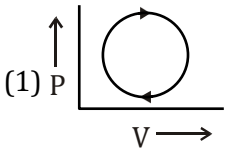
(2) $\frac{3}{2}$ PV only if the gas is diatomic.

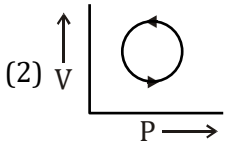
(3) $> \frac{3}{2}$ PV if the gas is diatomic.

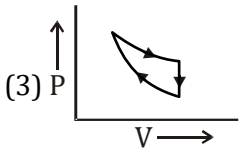
(4) $\frac{3}{2}$ PV in all cases.

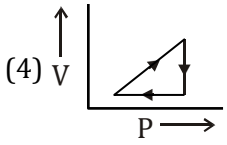
PTM517

12. The following are the P–V diagrams for cyclic processes for a gas. In which of these processes heat is released by the gas ?

(1) 

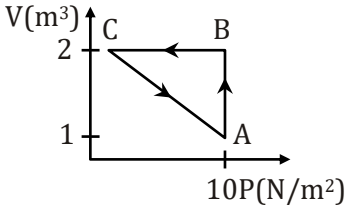
(2) 

(3) 

(4) 

PTM518

13. An ideal gas is taken through the cycle A → B → C → A, as shown in the figure. If the net heat supplied to the gas in the cycle is 5J, the work done by the gas in the process C → A is



(1) –5J (2) –10 J

(3) –15 J (4) –20 J

PTM519

14. The molar specific heat under constant pressure of oxygen is C_P = 7.03 cal/mol K. The quantity of heat required to raise the temperature from 10°C to 20°C of 5 moles of oxygen under constant volume will approximately be :-

(1) 25 cal (2) 50 cal

(3) 250 cal (4) 500 cal

PTM520

15. If the ratio of specific heat of a gas at constant pressure to that at constant volume is γ, the change in internal energy of a mass of gas, when the volume changes from V to 2 V at constant pressure P, is :-

(1) R/(γ – 1)

(2) PV

(3) PV/(γ – 1)

(4) γPV/(γ – 1)

PTM521

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	2	1	1	3	4	1	4	1	4	4	1	3	3

Exercise - II — Thermal Physics

1. Steam at 100°C is added slowly to 1400 g of water at 16°C until the temperature of water is raised to 80°C. The mass of steam required to do this is (L_V = 540 cal/g) :

(1) 160 g (2) 125 g

(3) 250 g (4) 320 g

PTM522

2. Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice of radius of P. The rate of fall of temperature of P is x times that of Q when both are at the same surface temperature. The value of x is :

(1) 1/4 (2) 1/3 (3) 3 (4) 4

PTM523

3. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of

(1) 4/3 (2) 16/9

(3) 64/27 (4) 256/81

PTM524

4. For a black body at temperature 727°C, its radiating power is 60 watt and temperature of surrounding is 227°C. If temperature of black body is changed to 1227°C then its radiating power will be :-

(1) 304 W (2) 320 W

(3) 240 W (4) 120 W

PTM525

5. An ideal gas expands in such a way that PV^2 = constant throughout the process.

(1) The graph of the process of T-V diagram is a parabola.

(2) The graph of the process of T-V diagram is a straight line.

(3) Such an expansion is possible only with heating.

(4) Such an expansion is possible only with cooling.

PTM526

6. N molecules of an ideal gas at temperature T_1 and pressure P_1 are contained in a closed box. If the molecules in the box gets doubled, Keeping total kinetic energy same. If new pressure is P_2 and temperature is T_2, Then:

(1) $P_2 = P, T_2 = T_1$

(2) $P_2 = P_1, T_2 = T_1 / 2$

(3) $P_2 = 2P_1, T_2 = T_1$

(4) $P_2 = 2P_1, T_2 = T_1 / 2$

PTM527

7. A gas mixture contain 1 g H_2 and 1 g He if temperature of gas mixture is increased from 0°C to 100° C at isobaric process. Then find given heat of gas mixture

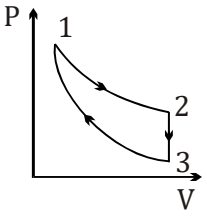
[$\gamma_{He} = 5/3, \gamma_{H_2} = 7/5$, R = 2 cal/mol-K]

(1) 124 cal (2) 327 cal

(3) 218 cal (4) 475 cal

PTM528

8. Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process 1 → 2 takes place at constant temperature (300K). Process 2 → 3 takes place at constant volume. During this process 40J of heat leaves the system. Process 3 → 1 is adiabatic and temperature T_3 is 275K. Work done by the gas during the process 3 → 1 is



(1) –40J (2) –20J (3) +40J (4) +20J

PTM529

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9. A bimetallic strip is formed out of two identical strips, one of copper and other of brass. The coefficients of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is

(1) Proportional to ΔT

(2) Proportional to $|\alpha_B - \alpha_C|$

(3) Inversely proportional to $|\alpha_B - \alpha_C|$

(4) none

PTM530

10. A cylinder of radius R made of material thermal conductivity K_1 is surrounded by a cylindrical shell of inner radius 2R made of material of thermal conductivity K_2. The two ends of combined system are maintained at two different temperature. There is no loss of heat & system is in steady state, calculate effective thermal conductivity of system: -

(1) $\frac{K_1 + 3K_2}{4}$ (2) $K_1 + K_2$

(3) $\frac{K_1 + 8K_2}{9}$ (4) $\frac{8K_1 + K_2}{9}$

PTM531

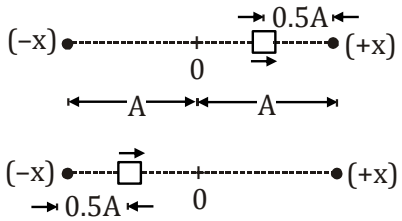
Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	3	4	2	4	2	4	1	3	1

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Exercise - I — Oscillations

1. Two bodies performing S.H.M. have same amplitude and frequency. Their phases at a certain instant are as shown in the figure. The phase difference between them is



(1) $\frac{11}{6}\pi$ (2) π (3) $\frac{\pi}{3}$ (4) $\frac{3}{5}\pi$

POS230

2. Two simple harmonic motions are represented by the equations $y_1 = 0.1 \sin\left(100\pi t + \frac{\pi}{3}\right)$ and $y_2 = 0.1 \cos 100\pi t$. The phase difference of the velocity of particle 1, with respect to the velocity of particle 2 is-

(1) $\frac{-\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{-\pi}{3}$ (4) $\frac{\pi}{6}$

POS231

3. The potential energy of a simple harmonic oscillator at mean position is 3 joules. If its mean K.E. is 4 joules, its total energy will be :

(1) 7J (2) 8J (3) 10J (4) 11J

POS232

4. The total energy of a harmonic oscillator of mass 2kg is 9 joules. If its potential energy at mean position is 5 joules, its K.E. at the mean position will be :

(1) 9J (2) 14J (3) 4J (4) 11J

POS233

5. A horizontal spring is connected to a mass M. It executes simple harmonic motion. When the mass M passes through its mean position, an object of mass m is put on it and the two move together. The ratio of frequencies before and after will be -

(1) $\left(1+\frac{m}{M}\right)^{1/2}$ (2) $\left(1+\frac{m}{M}\right)$

(3) $\left(\frac{M}{M+m}\right)^{1/2}$ (4) $\left(\frac{M}{M+m}\right)$

POS234

6. Two particles A and B of equal masses are suspended from two massless springs of spring constants k_1 and k_2, respectively. If the maximum velocities during oscillations are equal, the ratio of amplitudes of A and B is -

(1) $\sqrt{k_1/k_2}$ (2) k_1/k_2

(3) $\sqrt{k_2/k_1}$ (4) k_2/k_1

POS235

7. A particle describes SHM in a straight line about O.



If the time period of the motion is T then its kinetic energy at P be half of its peak value at O, if the time taken by the particle to travel from O to P is

(1) $\frac{1}{2}T$ (2) $\frac{1}{4}T$ (3) $\frac{1}{2\sqrt{2}}T$ (4) $\frac{1}{8}T$

POS236

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	3	1	4	3	1	3	4

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Exercise - II Oscillations

1. The period of a particle is 8s. At t = 0 it is at the mean position. The ratio of distance covered by the particle in first second and second will be-

(1) $\frac{\sqrt{2}-1}{\sqrt{2}}$ (2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{2}-1}$ (4) $\sqrt{2}-1$

POS237

2. A man of mass 60 kg standing on a platform executing S.H.M. in the vertical plane. The displacement from the mean position varies as y = 0.5 sin (2π ft). The minimum value of f, for which the man will feel weightlessness at the highest point is : (y is in metres)

(1) $\frac{g}{4\pi}$ (2) $4\pi g$

(3) $\frac{\sqrt{2g}}{2\pi}$ (4) $2\pi\sqrt{2g}$

POS238

3. A point mass oscillates along the x-axis according to the law $x = x_0\cos(\omega t - \pi/4)$. If the acceleration of the particle is written as- $a = A\cos(\omega t + \delta)$, then-

(1) $A = x_0, \delta = -\pi/4$ (2) $A = x_0\omega^2, \delta = \pi/4$

(3) $A = x_0\omega^2, \delta = -\pi/4$ (4) $A = x_0\omega^2, \delta = 3\pi/4$

POS239

4. A simple pendulum has time period T_1. The point of suspension is now moved upward according to the relation $y = Kt^2$, ($K = 1\ m/s^2$) where y is the vertical displacement. The time period now becomes T_2. The ratio of $\frac{T_1^2}{T_2^2}$ is :

($g = 10\ m/s^2$)

(1) $\frac{6}{5}$ (2) $\frac{5}{6}$

(3) 1 (4) $\frac{4}{5}$

POS240

5. An oscillator consists of a block attached to spring (k = 400 N/m), At some time t, the position (measured from the system's equilibrium location), velocity and acceleration of the block are x = 0.100m, v = – 15.0 m/s, and $a = -90\ m/s^2$. The amplitude of the motion and the mass of the block are :-

(1) 0.2 m, 0.84 kg

(2) 0.3 m, 0.76 kg

(3) 0.4 m, 0.54 kg

(4) 0.5 m, 0.44 kg

POS241

Exercise-II ANSWER KEY

Question	1	2	3	4	5
Answer	3	3	4	1	4

Exercise - I — Wave Motion

1. Transverse wave of same frequency are generated in two steel wires A and B. The diameter of A is twice of B and the tension in A is half that in B. The ratio of velocities of waves in A and B is –

(1) $1:\sqrt{2}$ (2) 1 : 2

(3) $3:2\sqrt{2}$ (4) $1:2\sqrt{2}$

PWA258

2. The power of sound from the speaker of a radio is 20 milliwatt. By turning the knob of the volume control the power of the sound is increased to 400 milliwatt. The power increase in decibles as compared to the original power is :

(1) 13 dB (2) 10 dB

(3) 20 dB (4) 800 dB

PWA259

3. A set of 56 tuning forks are so arranged in series that each fork gives 4 beats per second with the previous one. The freq. of the last fork is 3 times that of first. The freq. of first fork is :

(1) 110 (2) 60 (3) 56 (4) 65

PWA260

4. Two wires are fixed in a sonometer. Their tensions are in the ratio 8:1. The lengths are in the ratio 36:35. The diameters are in the ratio 4:1. Densities of the materials are in the ratio 1:2. If the higher frequency in the setting is 360 Hz, the beat frequency when the two wires sounded together is:

(1) 8 (2) 5 (3) 10 (4) 6

PWA261

5. Sound source of frequency 170 Hz is placed near a wall. A man walking from the source towards the wall finds, that there is periodic rise and fall of sound intensity. If the speed of sound in air is 340 m/s, then the distance separating the two adjacent portions of minimum intensity is:

(1) (1/2) m (2) (3/2) m

(3) 1 m (4) 2 m

PWA262

6. An open pipe is suddenly closed with the result that the second overtone of the closed pipe is found to be higher in frequency by 100 Hz, than the first overtone of the original pipe. The fundamental frequency of open pipe will be :

(1) 100 Hz (2) 300 Hz

(3)150 Hz (4) 200 Hz

PWA263

7. A man standing between two cliffs hears the first echo of a sound after 2 sec and the second echo 3 sec after the initial sound. If the speed of sound be 330 m/s, the distance between the two cliffs should be:

(1) 1650 m (2) 990 m

(3) 825 m (4) 660 m

PWA264

8. A vibrator makes 150 cm of a string to vibrate in 6 loops in the longitudinal arrangement when it is stretched by 150 N. The entire length of the string is then weighed and is found to weigh 400 mg. Then

(1) frequency of the vibrator is 3 kHz

(2) frequency of the vibrator is 1.5 kHz

(3) distance between two nodes is 25 cm

(4) distance between two nodes is 33 cm

PWA265

9. The equation of a plane progressive wave is $y = 0.02 \sin 8\pi\left[t-\frac{x}{20}\right]$. When it is reflected at a rarer medium (medium with higher velocity) at x = 0, its amplitude becomes 75% of its previous value. The equation of the reflected wave is

(1) $y=0.02\sin 8\pi\left[t-\frac{x}{20}\right]$

(2) $y=0.02\sin 8\pi\left[t+\frac{x}{20}\right]$

(3) $y=+0.015\sin 8\pi\left[t+\frac{x}{20}\right]$

(4) $y=-0.015\sin 8\pi\left[t+\frac{x}{20}\right]$

PWA266

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10. Length of a sonometer wire is either 95 cm or 100 cm, in both the cases a tuning fork produces 4 beats. Then the frequency of tuning fork is-

(1) 152 (2) 156

(3) 160 (4) 164

PWA267

11. A tuning fork gives 4 beats with 50 cm length of a sonometer wire. If the length of the wire is shortened by 1 cm then no. of beats still the same. The frequency of the fork is –

(1) 396 (2) 400

(3) 404 (4) 384

PWA268

12. In a resonance tube, the first resonance with a tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is :-

(1) 500 (2) 300

(3) 330 (4) 165

PWA269

13. An earthquake generates both transverse (S) and longitudinal (P) sound waves in the earth. The speed of S waves is about 4.5 km/s and that of P waves is about 8.0 km/s. A seismograph records P and S waves from an earthquake. The first P wave arrives 4.0 min before the first S wave. The epicenter of the earthquake is located at a distance of about

(1) 25 km

(2) 250 km

(3) 2500 km

(4) 5000 km

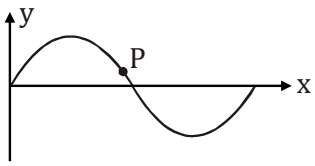
PWA270

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	4	1	1	3	3	4	3	2,3	3	2	1	1	3

Exercise - II — Wave Motion

1. A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. The wavelength of the wave is 0.5 m and its amplitude is 10 cm. At a particular time t, the snap shot of the wave is shown in figure. The velocity of P when its displacement is 5 cm is :



(1) $\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s (2) $-\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s

(3) $\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s (4) $-\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s

PWA271

2. Frequency of tuning fork A is 256 Hz. It produces four beats/sec with tuning fork B. When wax is applied at tuning fork B then 6 beats/sec are heard. Frequency of B is :

(1) 252

(2) 260 Hz

(3) (1) & (2) both

(4) 264

PWA272

3. An under water swimmer sends a sound signal to the surface. It produces 5 beats/sec when compared with fundamental tone of a pipe of 20 cm length closed at one end. What is wavelength of sound in water.

(take V_{water} = 1500 m/sec, V_{air} = 360m/sec)

(1) 3.3 m or 3.37 m (2) 4.4 m or 4.47 m

(3) 2.5 m or 2.7 m (4) 1m or 1.7 m

PWA273

4. A cylindrical tube (L = 120 cm.) is resonant with a tuning fork of frequency 330 Hz. If it is filling by water then to get resonance minimum length of water column is (V_{air} = 330 m/s)

(1) 45 cm. (2) 60 cm.

(3) 25 cm. (4) 20 cm.

PWA274

5. A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges when a mass of 9kg is suspended from the wire. When this mass is replaced by mass M, the wire resonates with the same tuning fork forming three antinodes for the same positions of the bridges. Then find the value of square root of M.

(1) 5 (2) 10 (3) 25 (4) None

PWA275

Exercise-II — ANSWER KEY

Question	1	2	3	4	5
Answer	3	3	1	1	1